US 20250285950A1

(19) United States
(12) Patent Application Publication
LEE

(10) Pub. No.: US 2025/0285950 A1
(43) Pub. Date: Sep. 11, 2025

(54) **SEMICONDUCTOR DEVICE PACKAGE**

(71) Applicant: **Advanced Semiconductor Engineering, Inc.**, Kaohsiung (TW)

(72) Inventor: **Yu-Ying LEE**, Kaohsiung (TW)

(73) Assignee: **Advanced Semiconductor Engineering, Inc.**, Kaohsiung (TW)

(21) Appl. No.: **19/220,061**

(22) Filed: **May 27, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/963,959, filed on Oct. 11, 2022, now Pat. No. 12,315,785, which is a continuation of application No. 16/932,690, filed on Jul. 17, 2020, now Pat. No. 11,469,165, which is a continuation of application No. 15/858,939, filed on Dec. 29, 2017, now Pat. No. 10,741,482.

Publication Classification

(51) **Int. Cl.**

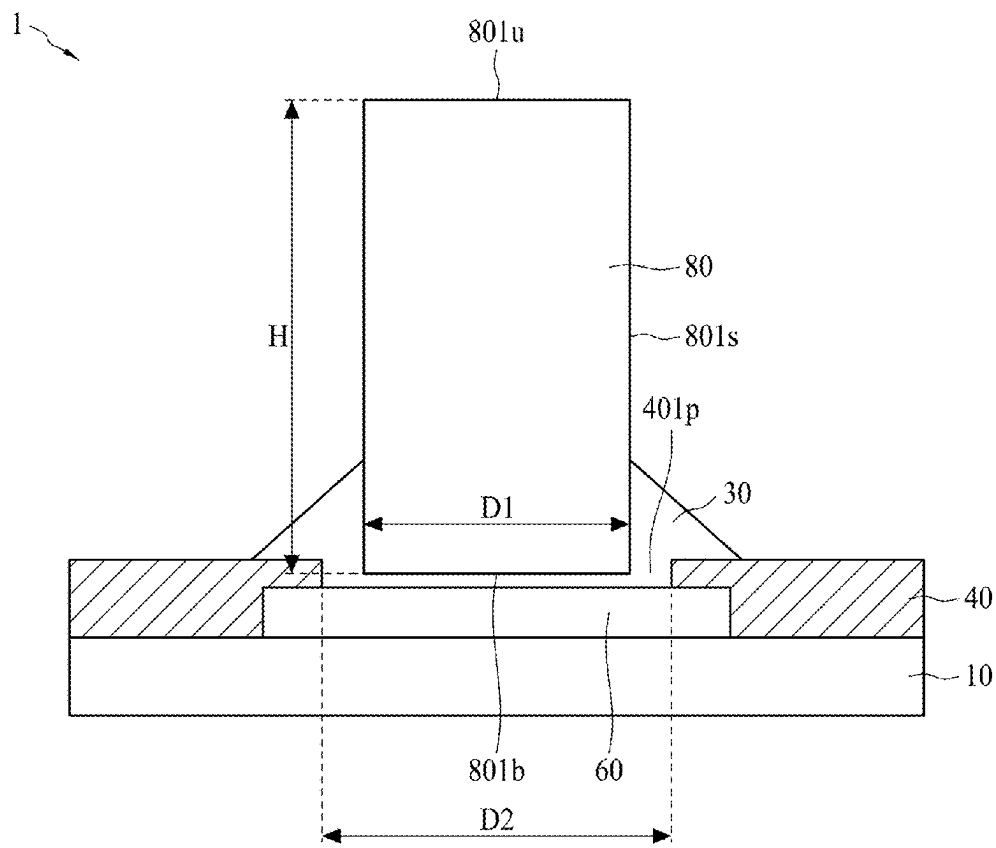
H01L 23/498	(2006.01)
H01L 21/48	(2006.01)
H01L 23/00	(2006.01)
H01L 23/31	(2006.01)
H01L 23/495	(2006.01)
H01L 25/10	(2006.01)

(52) **U.S. Cl.**

CPC ***H01L 23/49811*** (2013.01); ***H01L 21/4853*** (2013.01); ***H01L 21/4857*** (2013.01); ***H01L 23/3128*** (2013.01); ***H01L 23/49517*** (2013.01); ***H01L 23/49822*** (2013.01); ***H01L 23/49866*** (2013.01); ***H01L 24/16*** (2013.01); ***H01L 24/29*** (2013.01); ***H01L 24/32*** (2013.01); *H01L 23/49816* (2013.01); *H01L 24/13* (2013.01); *H01L 25/10* (2013.01); *H01L 2224/13147* (2013.01); *H01L 2224/16225* (2013.01); *H01L 2224/16237* (2013.01); *H01L 2224/29298* (2013.01); *H01L 2224/32227* (2013.01); *H01L 2224/8185* (2013.01); *H01L 2225/1058* (2013.01); *H01L 2924/15311* (2013.01); *H01L 2924/181* (2013.01); *H01L 2924/19105* (2013.01)

(57) **ABSTRACT**

A semiconductor device package includes a carrier, a first conductive post and a first adhesive layer. The first conductive post is disposed on the carrier. The first conductive post includes a lower surface facing the carrier, an upper surface opposite to the lower surface and a lateral surface extended between the upper surface and the lower surface. The first adhesive layer surrounds a portion of the lateral surface of the first conductive post. The first adhesive layer comprises conductive particles and an adhesive. The first conductive post has a height measured from the upper surface to the lower surface and a width. The height is greater than the width.



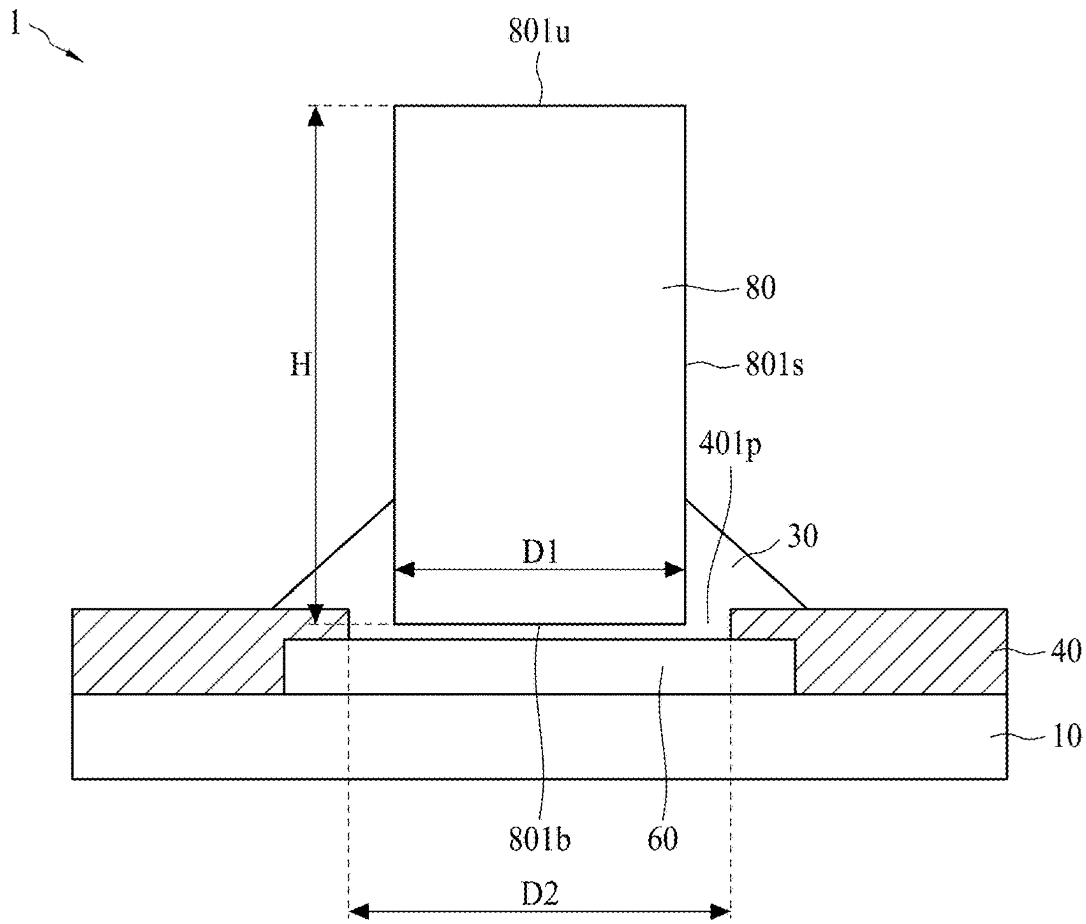

1
801u
80
801s
H
401p
30
D1
40
10
801b
60
D2

FIG. 1

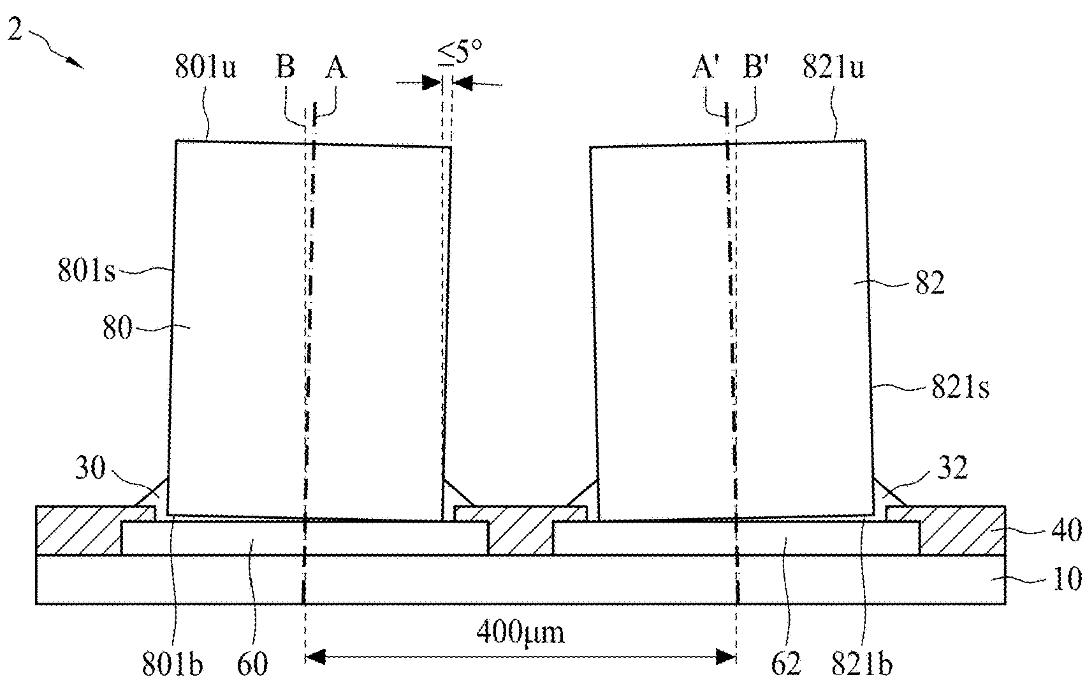

2
801u
B
A
≤5°
A'
B'
821u
801s
80
82
821s
30
32
40
10
400μm
801b
60
62
821b

FIG. 2A

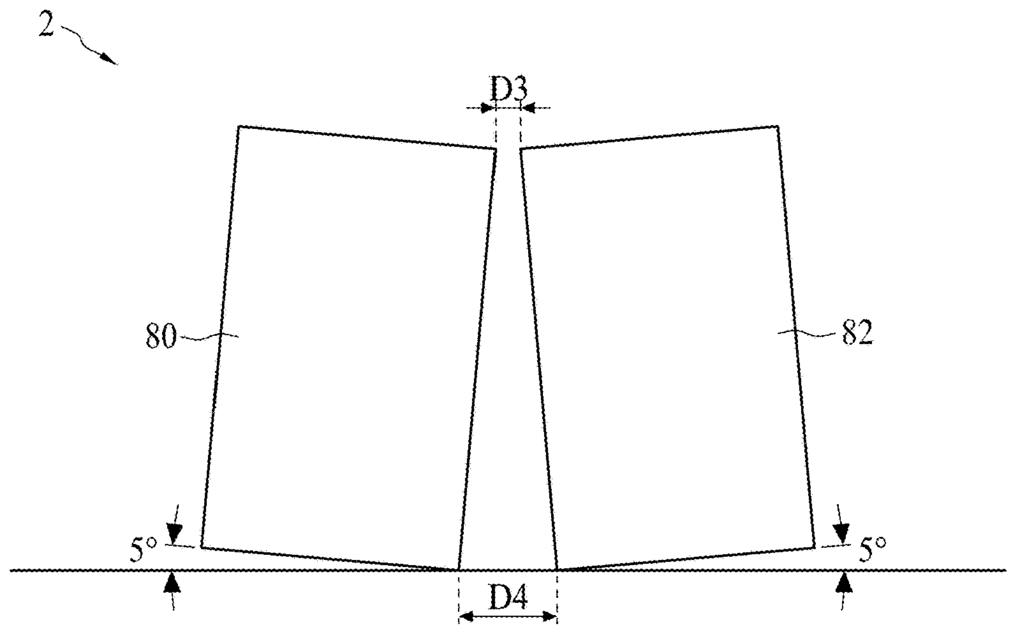


FIG. 2B

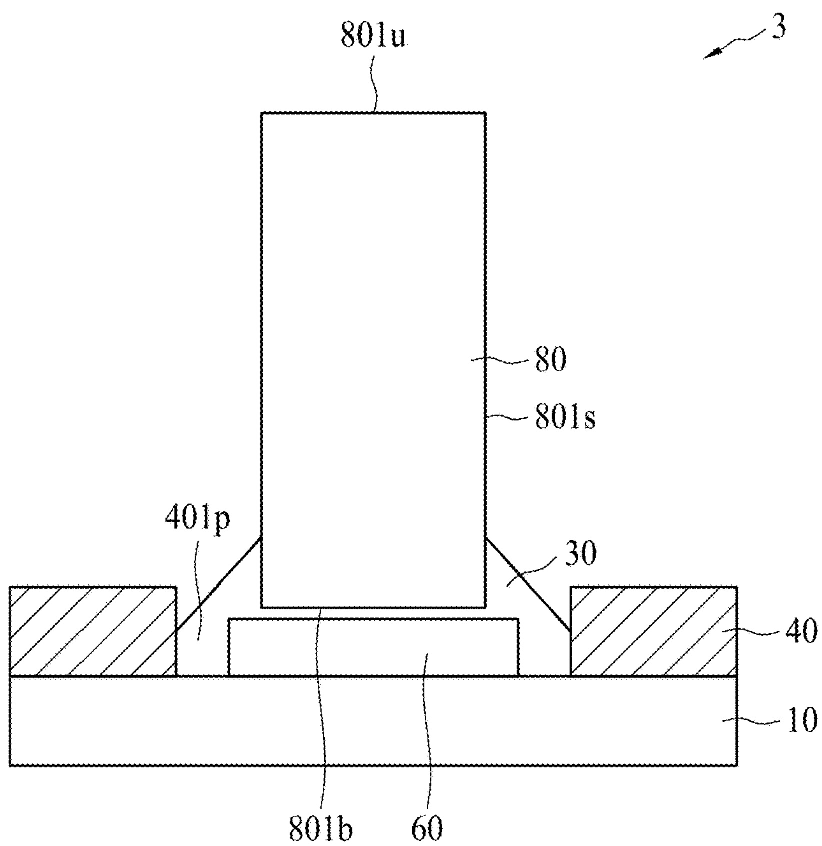


FIG. 3

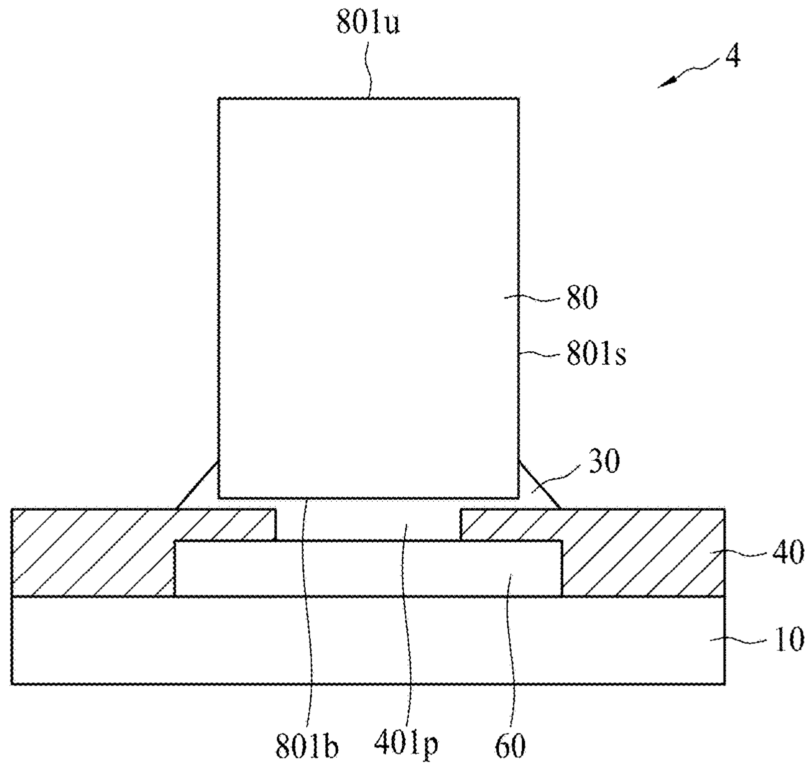

801u
4
80
801s
30
40
10
801b
401p
60

FIG. 4

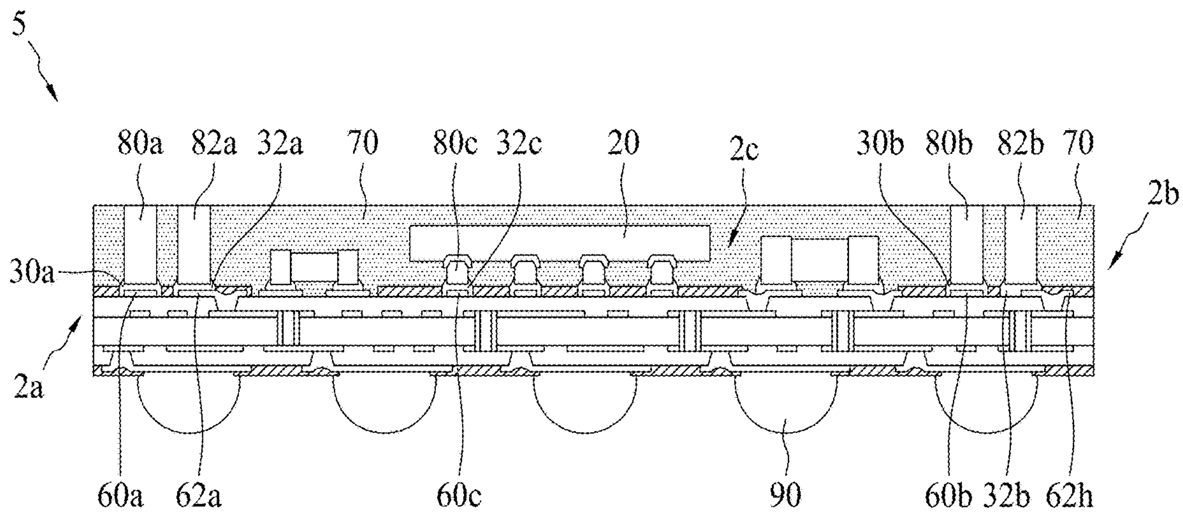

5
80a
82a
32a
70
80c
32c
20
2c
30b
80b
82b
70
2b
30a
2a
60a
62a
60c
90
60b
32b
62h

FIG. 5

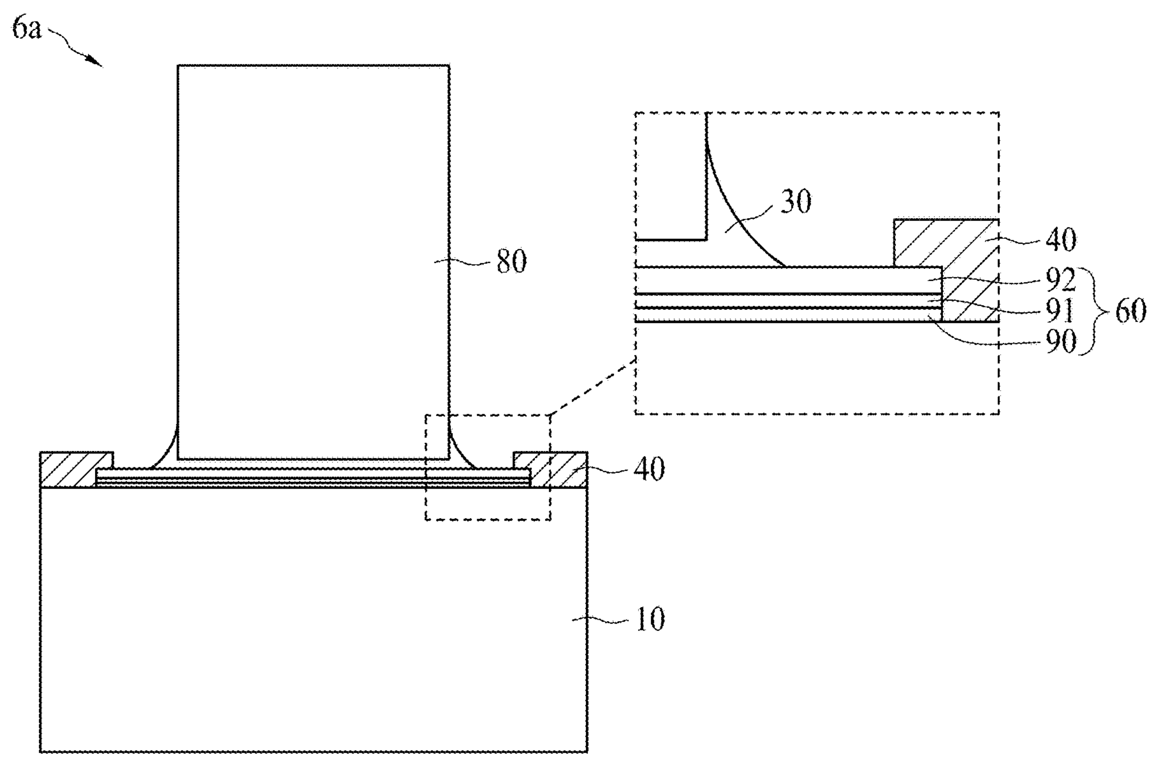


FIG. 6A

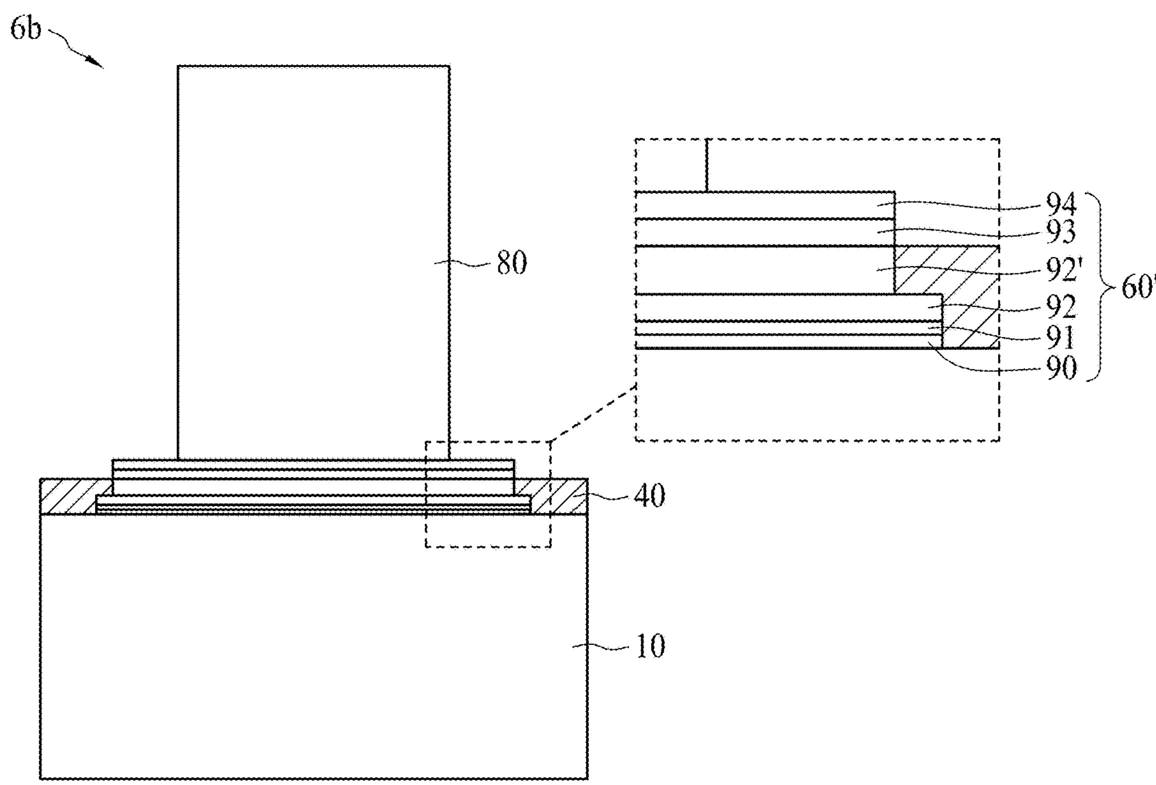


FIG. 6B

FIG. 7A

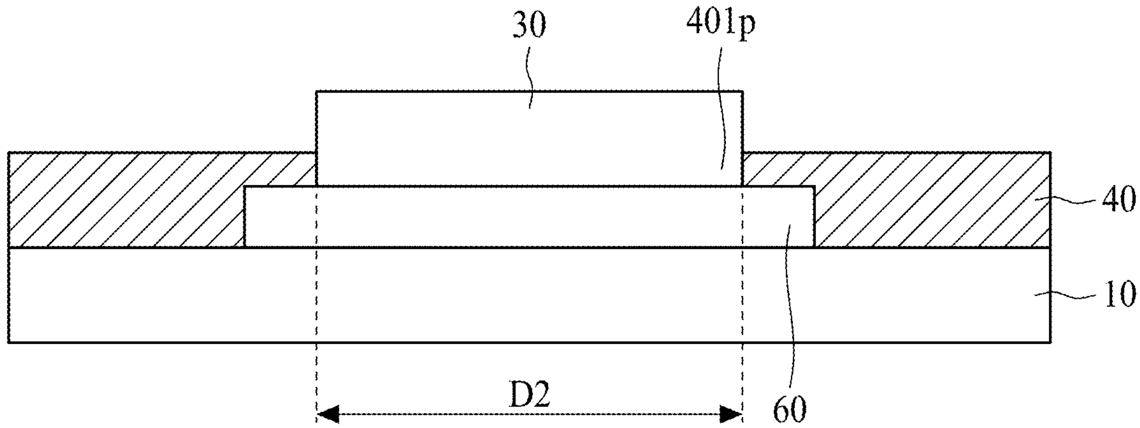


FIG. 7B

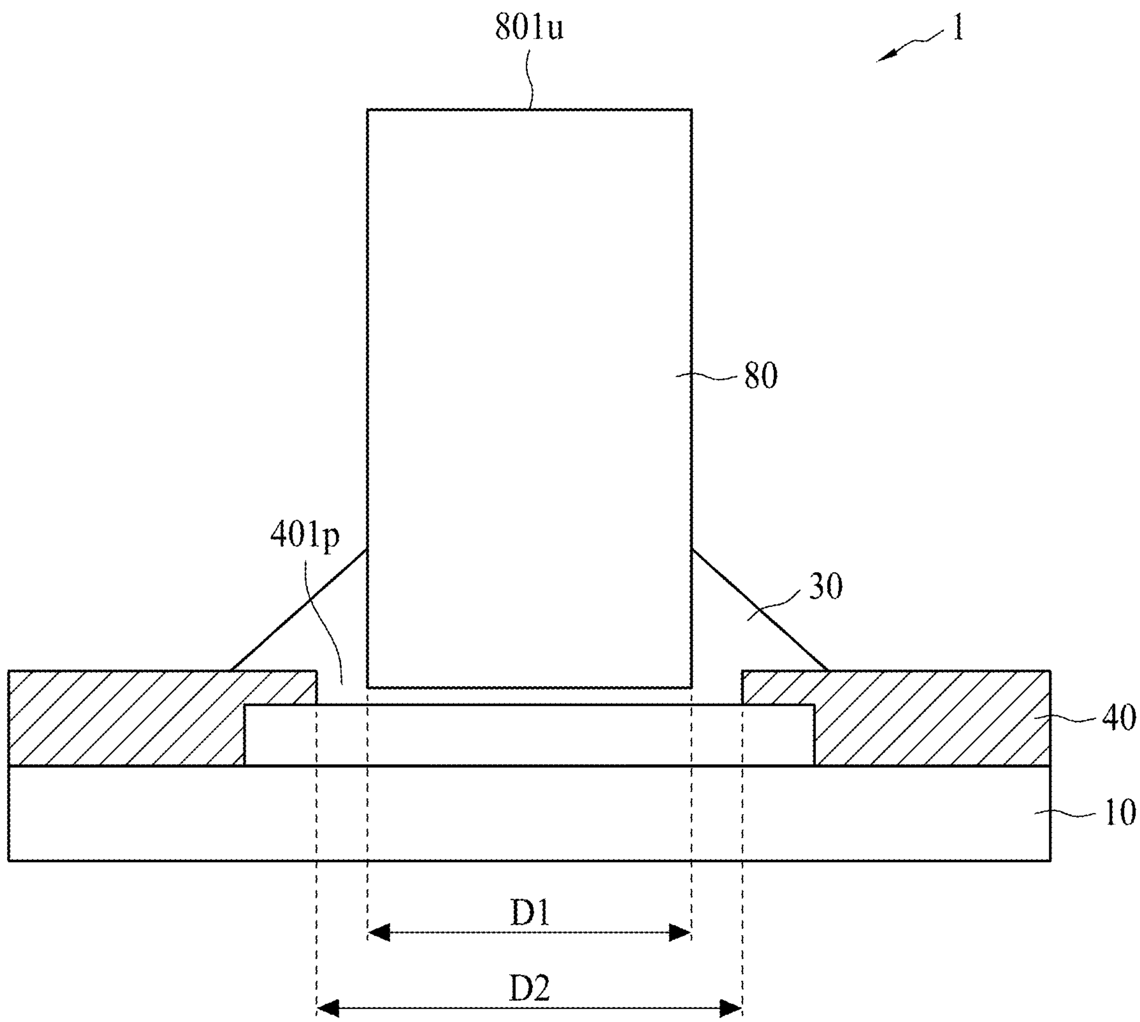


FIG. 7C

SEMICONDUCTOR DEVICE PACKAGE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 17/963,959 filed Oct. 11, 2022, now issued as U.S. Pat. No. 12,315,785 which is a continuation of U.S. patent application Ser. No. 16/932,690 filed Jul. 17, 2020, now issued as U.S. Pat. No. 11,469,165 which is a continuation of U.S. patent application Ser. No. 15/858,939 filed Dec. 29, 2017, now issued as U.S. Pat. No. 10,741,482 the contents of each of which are incorporated herein by reference in their entireties.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a semiconductor device package and a method of manufacturing the same.

2. Description of the Related Art

[0003] A conductive post in a semiconductor device package may move, tilt or fall because solder material between the conductive post and a conductive pad (of a substrate) is melted during thermal cycles in a process of manufacturing the semiconductor device package. The moved, tilted or fell conductive post may cause reliability issues of the semiconductor device package.

SUMMARY

[0004] In one or more embodiments, a semiconductor device package includes a carrier, a first conductive post and a first adhesive layer. The first conductive post is disposed on the carrier. The first conductive post includes a lower surface facing the carrier, an upper surface opposite to the lower surface and a lateral surface extended between the upper surface and the lower surface. The first adhesive layer surrounds a portion of the lateral surface of the first conductive post. The first adhesive layer comprises conductive particles and an adhesive. The first conductive post has a height measured from the upper surface to the lower surface and a width. The height is greater than the width.

[0005] In one or more embodiments, a semiconductor device package includes a carrier, a conductive post, a first adhesive layer and an insulating layer. The conductive post is disposed on the carrier and including a lateral surface. The first adhesive layer surrounds a portion of the lateral surface of the conductive post. The first adhesive layer includes conductive particles and an adhesive. The insulating layer encapsulates the conductive post and exposes an external contact of the conductive post.

[0006] In one or more embodiments, a method for manufacturing a semiconductor device includes disposing an adhesive layer comprising conductive particles and an adhesive on a carrier; disposing a pretreated or preformed conductive post on the adhesive layer; and curing the adhesive layer.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a cross-sectional view of a semiconductor device package in accordance with some embodiments of the present disclosure.

[0008] FIG. **2**A is a cross-sectional view of a semiconductor device package in accordance with some embodiments of the present disclosure.

[0009] FIG. **2**B is a schematic diagram of a cross-sectional view of the semiconductor device package of FIG. **2**B.

[0010] FIG. **3** is a cross-sectional view of a semiconductor device package in accordance with some embodiments of the present disclosure.

[0011] FIG. **4** is a cross-sectional view of a semiconductor device package in accordance with some embodiments of the present disclosure.

[0012] FIG. **5** is a cross-sectional view of a semiconductor device package in accordance with some embodiments of the present disclosure.

[0013] FIG. **6**A is a cross-sectional view of a semiconductor device package in accordance with some embodiments of the present disclosure.

[0014] FIG. **6**B is a cross-sectional view of a semiconductor device package in accordance with some embodiments of the present disclosure.

[0015] FIG. **7**A illustrates one or more stages of a method of manufacturing a semiconductor device package in accordance with some embodiments of the present disclosure.

[0016] FIG. **7**B illustrates one or more stages of a method of manufacturing a semiconductor device package in accordance with some embodiments of the present disclosure.

[0017] FIG. **7**C illustrates one or more stages of a method of manufacturing a semiconductor device package in accordance with some embodiments of the present disclosure.

[0018] Common reference numerals are used throughout the drawings and the detailed description to indicate the same or similar elements. Embodiments of the present disclosure will be more apparent from the following detailed description taken in conjunction with the accompanying drawings.

DETAILED DESCRIPTION

[0019] Spatial descriptions, such as "above," "below," "up," "left," "right," "down," "top," "bottom," "vertical," "horizontal," "side," "higher," "lower," "upper," "over," "under," and so forth, are indicated with respect to the orientation shown in the figures unless otherwise specified. It should be understood that the spatial descriptions used herein are for purposes of illustration, and that practical implementations of the structures described herein can be spatially arranged in any orientation or manner, provided that the merits of embodiments of this disclosure are not deviated by such arrangement.

[0020] FIG. **1** is a cross-sectional view of a semiconductor device package **1** in accordance with some embodiments of the present disclosure. The semiconductor device package **1** includes a carrier **10**, an adhesive layer **30**, a dielectric layer **40**, a pad **60** and a conductive post **80**. The adhesive layer **30** may include, e.g., conductive glue.

[0021] The conductive post **80** is disposed on the carrier **10**. The conductive post **80** may be, or include, a pre-formed conductive post. The conductive post **80** includes a lower surface **801***b* facing the carrier **10**, an upper surface **801***u* opposite to the lower surface **801***b* and a lateral surface **801***s* extended between the upper surface **801***u* and the lower surface **801***b*. In some embodiments, the dielectric layer **40** includes Polypropylene (PP), Polyimide (PI), Ajinomoto Build-up Film (ABF), other suitable insulating materials, or a combination of two or more thereof.

[0022] In some embodiments, the carrier **10** includes silicon (Si), ceramic, glass, metal, other suitable inorganic materials or a combination of two or more thereof. The carrier **10** may be a substrate, a wafer, or a lead frame. In some embodiments, each of the pad **60** and the conductive post **80** includes, for example, copper (Cu), other metal, a metal alloy, other conductive material, or a combination of two or more thereof. The adhesive layer **30** surrounds a portion of the lateral surface **801***s* of the conductive post **80**. In some embodiments, the adhesive layer **30** includes a conductive filler such as conductive particles (such as gold (Au), silver (Ag), Cu, another metal, a metal alloy, or other conductive material) and an adhesive (such as epoxy resin or other resin).

[0023] In some embodiments, a thermal degradation temperature (Td) of the adhesive of the adhesive layer **30** may be higher than a reflow temperature in a reflow process. A melting temperature of the adhesive of the adhesive layer **30** may be higher than the reflow temperature in the reflow process. For example, the melting temperature of the adhesive layer **30** may higher than about 260 Degrees Celsius (° C.). Thus, the adhesive layer **30** may not be melted or reflowed at a working temperature ranging from about 25° C. to about 260° C. during a reflow process. At least a portion of the lateral surface **801***s* of the conductive post **80** may be surrounded by the adhesive layer **30**, which has a relatively high melting point, may fix or support the conductive post **80** during thermal cycles in a process of manufacturing the semiconductor device package **1**.

[0024] The conductive post **80** has a height H measured from the upper surface **801***u* to the lower surface **801***b*. In some embodiments, the height H ranges from about 410 micrometers (μm) to about 490 μm. The conductive post **80** has a width D**1**. In some embodiments, the width D**1** ranges from about 270 μm to about 330 μm. The height H of the conductive post **80** is greater than the width D**1** of the conductive post **80**. The roughness of the lower surface **801***b* may be different from the lateral surface **801***s*. In some embodiments, the roughness of the lower surface **801***b* is greater than the lateral surface **801***s*. The ratio of the height H of the conductive post **80** to the width D**1** of the conductive post **80** may be equal to or more than about 1.2:1, such as about 1.3:1 or greater, about 1.4:1 or greater, or about 1.5:1 or greater. The relatively great roughness of the lower surface **801***b* of the conductive post **80** may enhance the adhesion and conductivity between the lower surface **801***b* and the adhesive layer **30**. In some embodiments, a relatively larger sized conductive filler (e.g., conductive particles) of the adhesive layer **30** is in contact with the lower surface **801***b* and the pad **60**. A relatively smaller sized conductive filler of the adhesive layer **30** may be fitted into the uneven surface **801***b* of the conductive post **80** to mitigate a void issue at the lower surface **801***b* of the conductive post **80**.

[0025] The pad **60** is disposed between the carrier **10** and the conductive post **80**. The dielectric layer **40** and the pad **60** are disposed on the carrier **10**. The dielectric layer **40** defines an opening **401***p* exposing the pad **60**. The pad **60** can be a conductive pad. The conductive post **80** is electrically connected to the pad **60** through the adhesive layer **30**. The conductive post **80** is electrically connected to the pad **60** by the adhesive layer **30** in the opening **401***p*. In some embodiments, the conductive post **80** is not in direct contact with the pad **60**, with a gap in between. In some other embodiments, at least a portion of the lower surface **801***b* is in direct contact with the pad **60**.

[0026] A difference between a length D**2** of the opening **401***p* of the dielectric layer **40** and a width D**1** of the conductive post **80** may be equal to or greater than about 70 μm. The lower surface **801***b* of the conductive post **80** is disposed within the opening **401***p*. A portion of the conductive post **80** may be disposed on the dielectric layer **40**. The difference between the length D**2** of the opening **401***p* of the dielectric layer **40** and the width D**1** of the conductive post **80** may be greater than about 70 μm so that a portion of the lateral surface **801***s* of the conductive post **80** can be surrounded and supported by the adhesive layer **30**.

[0027] In some embodiments, one or more conductive posts may be slightly tilted with respect to a surface normal of a carrier. FIG. **2**A is a cross-sectional view of a semiconductor device package **2** in accordance with some embodiments of the present disclosure. The semiconductor device package **2** includes a carrier **10**, adhesive layers **30** and **32**, a dielectric layer **40**, pads **60** and **62** and conductive posts **80** and **82**. The adhesive layers **30** may include, e.g., conductive glues.

[0028] The conductive posts **80** and **82** are disposed on the carrier **10**. The conductive post **80** includes a lower surface **801***b* facing the carrier **10**, an upper surface **801***u* opposite to the lower surface **801***b* and a lateral surface **801***s* extended between the upper surface **801***u* and the lower surface **801***b*. Similarly, the conductive post **82** includes a lower surface **821***b* facing the carrier **10**, an upper surface **821***u* opposite to the lower surface **821***b* and a lateral surface **821***s* extended between the upper surface **821***u* and the lower surface **821***b*. In some embodiments, the dielectric layer **40** includes PP, PI, ABF, other suitable insulating materials or a combination of two or more thereof. In some embodiments, the carrier **10** includes Si ceramic, glass, metal, other suitable inorganic materials, or a combination of two or more thereof. In some embodiments, each of the pads **60** and **62** and conductive posts **80** and **82** includes for example, Cu, other metal, a metal alloy, other conductive material, or a combination of two or more thereof.

[0029] The adhesive layer **30** surrounds a portion of the lateral surface **801***s* of the conductive post **80**. The adhesive layer **32** surrounds a portion of the lateral surface **821***s* of the conductive post **82**. In some embodiments, the adhesive layer **30** includes a conductive filler such as conductive particles (such as Au, Ag or Cu) and an adhesive (such as epoxy resin). At least a portion of the lateral surface **801***s* of the conductive post **80** may be surrounded by the adhesive layer **30**, which has a relatively high melting point, and may fix or support the conductive post **80** during thermal cycles in a process of manufacturing the semiconductor device package **2**. Similarly, at least a portion of the lateral surface **821***s* of the conductive post **82** may be surrounded by the adhesive layer **32**, which has a relatively high melting point, and may fix or support the conductive post **82** during the thermal cycles in the process of manufacturing the semiconductor device package **2**.

[0030] As shown in FIG. **2**A, the conductive post **80** has a vertical geometrical central axis A and the pad **60** has a vertical geometrical central axis B. The vertical geometrical central axis B of the pad **60** may be parallel to a surface normal of the carrier **10**. The conductive post **82** has a vertical geometrical central axis A' and the pad **62** has a vertical geometrical central axis B'. The vertical geometrical central axis B' of the pad **62** may be parallel to the surface normal of the carrier **10**. In some embodiments, a distance (also referred to as pitch) between the vertical geometrical central axis B of the pad **60** and the vertical geometrical central axis B' of the pad **62** may be about 400 μm.

[0031] A tilt angle between the vertical geometrical central axis A of the conductive post **80** and the vertical geometrical central axis B of the pad **60** may be equal to or smaller than about 5 degrees, such as about 4 degrees or less, or about 3 degrees or less. A tilt angle between the vertical geometrical central axis A' of the conductive post **82** and the vertical geometrical central axis B' of the pad **62** may be equal to or smaller than about 5 degrees, such as about 4 degrees or less, or about 3 degrees or less. A tilt angle defined by the vertical geometrical central axis A of the conductive post **80** relative to the vertical geometrical central axis B of the pad **60** may be equal or smaller than about 5 degrees and may help to avoid bridge (short circuit) of the adjacent two conductive posts **80** and **82**.

[0032] FIG. **2**B is a schematic diagram of a cross-sectional view of a semiconductor device package **2** of FIG. **2**A. In some embodiments, the minimum distance D**3** between the two conductive posts **80** and **82** of the semiconductor device package **2** may be about 21.56 μm. The distance D**4** between the bottom of the two conductive posts **80** and **82** of the semiconductor device package **2** may be about 100 μm.

[0033] FIG. **3** is a cross-sectional view of a semiconductor device package **3** in accordance with some embodiments of the present disclosure. The semiconductor device package **3** is similar to the semiconductor device package **1** of FIG. **1**, except that an adhesive layer **30** is not disposed on a top surface of a dielectric layer **40**. Some of the same-numbered components are not described again with respect to FIG. **3**. The semiconductor device package **3** includes a carrier **10**, the adhesive layer **30**, the dielectric layer **40**, a pad **60** and a conductive post **80**.

[0034] In some embodiments, the adhesive layer **30** includes a conductive filler such as conductive particles (such as Au, Ag or Cu) and an adhesive (such as epoxy resin). A thermal degradation temperature (Td) of the adhesive of the adhesive layer **30** may be higher than a reflow temperature in a reflow process. A melting temperature of the adhesive of the adhesive layer **30** may be higher than the reflow temperature in the reflow process. For example, the melting temperature of the adhesive layer **30** may be greater than about 260° C. The pad **60** is disposed between the carrier **10** and the conductive post **80**. The dielectric layer **40** and the pad **60** are disposed on the carrier **10**. The dielectric layer **40** defines an opening **401***p* exposing the pad **60**. The conductive post **80** is electrically connected to the pad **60** through the adhesive layer **30**. The conductive post **80** is electrically connected to the pad **60** by the adhesive layer **30** in the opening **401***p*.

[0035] A portion of the lateral surface **801***s* which is extended between the upper surface **801***u* and the lower surface **801***b* may be supported by the adhesive layer **30**. The adhesive layer **30** is not disposed on the top surface of the dielectric layer **40**. The adhesive layer **30** contacts a portion of a side wall of the opening **401***p* of the dielectric layer **40**.

[0036] FIG. **4** is a cross-sectional view of a semiconductor device package **4** in accordance with some embodiments of the present disclosure. The semiconductor device package **4** is similar to the semiconductor device package **1** of FIG. **1**, except that a portion of an adhesive layer **30** is disposed on a top surface of a dielectric layer **40**. At least some of same-numbered components are not described again with respect to FIG. **4**. The semiconductor device package **4** includes a carrier **10**, the adhesive layer **30**, the dielectric layer **40**, a pad **60** and a conductive post **80**.

[0037] The conductive post **80** is disposed on the carrier **10** and the dielectric layer **40**. The conductive post **80** includes a lower surface **801***b* facing the carrier **10**, an upper surface **801***u* opposite to the lower surface **801***b* and a lateral surface **801***s* extended between the upper surface **801***u* and the lower surface **801***b*. The adhesive layer **30** surrounds a portion of the lateral surface **801***s* of the conductive post **80**. A portion of the adhesive layer **30** is disposed on the top surface of the dielectric layer **40**.

[0038] The dielectric layer **40** defines an opening **401***p* exposing the pad **60**. The conductive post **80** is electrically connected to the pad **60** through the adhesive layer **30**. The conductive post **80** is disposed above the top surface of the dielectric layer **40**. The conductive post **80** is electrically connected to the pad **60** by the adhesive layer **30** in the opening **401***p*.

[0039] FIG. **5** is a cross-sectional view of a semiconductor device package **5** in accordance with some embodiments of the present disclosure. The semiconductor device package **5** includes a semiconductor device package **2***a*, a semiconductor device package **2***b*, a semiconductor device package **2***c*, an electronic component **20**, an insulating layer **70** and solder bumps **90**.

[0040] The semiconductor device packages **2***a*, **2***b* and **2***c* are similar to the semiconductor device package **1** of FIG. **1**, and at least some of the same-numbered components are not described again with respect to FIG. **5**. The semiconductor device package **2***a* includes conductive posts **80***a* and **82***a*, adhesive layers **30***a* and **32***a*, a dielectric layer **40**, and pads **60***a* and **62***a*. The semiconductor device package **2***b* includes conductive posts **80***b* and **82***b*, adhesive layers **30***b* and **32***b*, a dielectric layer **40**, and pads **60***b* and **62***b*. The semiconductor device package **2***c* includes conductive posts **80***c*, an adhesive layer **32***c*, a dielectric layer **40**, and a pad **60***c*.

[0041] The conductive post **80***c* may be disposed below the electronic component **20**. The conductive post **80***c* is supported by the adhesive layer **32***c*. The adhesive layer **32***c* includes a conductive filler and an insulating material. The electronic component **20** is disposed on the carrier **10**. The electronic component **10** is adjacent to the conductive posts **80***a*, **82***a*, **80***b* and **82***b*. The insulating layer **70** encapsulates the electronic component **20** and the conductive posts **80***a*, **82***a*, **80***b* and **82***b*. For example, the insulating layer **70** encapsulates the conductive posts **80***a*, **80***b*, **82***a* and **82***b* and exposes the top surfaces of external contacts of the conductive posts **80***a*, **80***b*, **82***a* and **82***b*. In some embodiments, the roughness of a lower surface of the conductive post **80***a* corresponding to the adhesive layer **30***a* is different from the roughness of the lateral surface of the conductive post **80***a*. The roughness of the lower surface of the conductive post **80***a* may be greater than the roughness of the lateral surface conductive post **80***a*. In some embodiments, a tilt angle between a vertical geometrical central axis of the conductive post **80***c* and a vertical geometrical central axis of the pad **60***c* may be equal to or smaller than 5 degrees.

[0042] FIG. **6**A is a cross-sectional view of a semiconductor device package **6***a* in accordance with some embodiments of the present disclosure. The semiconductor device package **6***a* is similar to the semiconductor device package

1 of FIG. **1**, and some of the same-numbered components are not described again with respect to FIG. **6**. The semiconductor device package **6***a* includes a carrier **10**, an adhesive layer **30**, a dielectric layer **40**, a pad **60** and a conductive post **80**.

[0043] In some embodiments, the pad **60** includes layers **90**, **91** and **92**. In some embodiments, the layer **90** may include, for example, aluminum (Al), other metal, a metal alloy, other conductive material, or a combination of two or more thereof. In some embodiments, the layer **91** may include, for example, titanium (Ti), other metal, r a metal alloy, other conductive material, or a combination of two or more thereof. In some embodiments, the layer **92** may include, for example, Cu, other metal, a metal alloy, other conductive material, or a combination of two or more thereof. In some embodiments, the adhesive layer **30** includes a conductive filler such as conductive particles (such as Au, Ag or Cu) and an adhesive (such as epoxy resin). A portion of the lateral surface **801***s* of the conductive post **80** is supported by the adhesive layer **30**. A portion of the lateral surface **801***s* of the conductive post **80** is surrounded by the adhesive layer **30**, which have a relatively high melting point, and may fix or support conductive post **80** during thermal cycles in a process of manufacturing the semiconductor device package **6***a*.

[0044] FIG. **6**B is a cross-sectional view of a semiconductor device package **6***b* in accordance with some embodiments of the present disclosure. The semiconductor device package **6***b* is similar to the semiconductor device package **1** of FIG. **1**, and some of same-numbered components are not described again with respect to FIG. **6**B. The semiconductor device package **6***b* includes a carrier **10**, a dielectric layer **40**, a pad **60'** and a conductive post **80**.

[0045] In some embodiments, the pad **60'** includes layers **90**, **91**, **92**, **92'**, **93** and **94**. In some embodiments, the layer **90** may include, for example, Al, or other metal, a metal alloy, other conductive material, or a combination of two or more thereof. In some embodiments, the layer **91** may include, for example, Ti, or other metal, a metal alloy, other conductive material, or a combination of two or more thereof. In some embodiments, the layer **92** may be a seed layer including, for example, Cu, or other metal, a metal alloy, other conductive material, or a combination of two or more thereof. In some embodiments, the layer **92'** may include, for example, Cu, or other metal, a metal alloy, other conductive material, or a combination of two or more thereof.

[0046] In some embodiments, the layer **93** may include, for example, nickel (Ni), or other metal, a metal alloy, other conductive material, or a combination of two or more thereof. In some embodiments, the layer **93** may include, for example, tin (Sn), or other metal, a metal alloy, other conductive material, or a combination of two or more thereof. The layer **93** of Sn has a relatively low melting point. The conductive post **80** may be tilted during thermal cycles in a process of manufacturing the semiconductor device package **6***b* since the layer **93** may be reflowed.

[0047] FIGS. **7**A-**7**C illustrate various stages of a method of manufacturing the semiconductor device package **1**. Referring to FIG. **7**A, a carrier **10** is provided. In some embodiments, the carrier **10** includes Si, ceramic, glass, metal, other suitable inorganic materials, or a combination of two or more thereof.

[0048] Referring to FIG. **7**B, a pad **60** is disposed on the top surface of the carrier **10**. In some embodiments, the pad **60** include, for example, Cu, or other metal, a metal alloy, other conductive material, or a combination of two or more thereof. A dielectric layer **40** is disposed on the top surface of the carrier **10** and covers at least a portion of the pad **60**. In some embodiments, the dielectric layer **40** includes PP, PI, ABF, other suitable insulating materials, or a combination of two or more thereof. The dielectric layer **40** defines an opening **401***p* exposing the pad **60**. The opening **401***p* of the dielectric layer **40** has a length D**2**. An adhesive layer **30** is disposed in the opening **401***p* and protrudes from the top surface of the dielectric layer **40**. In some embodiments, the adhesive layer **30** includes a conductive filler such as conductive particles (such as Au, Ag or Cu) and an adhesive (such as epoxy resin). The disposing of the adhesive layer may include providing a stencil to print the adhesive layer **30** on the pad **60** of the carrier **10** with a pattern defined by the stencil. The stencil may include an opening to align the pretreated conductive post **80** to connect the adhesive layer **30** through the opening of the stencil.

[0049] A thermal degradation temperature (Td) of the adhesive of the adhesive layer **30** may be higher than a reflow temperature in a reflow process. A melting temperature of the adhesive of the adhesive layer **30** of the adhesive layer **30** may be higher than the reflow temperature in the reflow process. In some embodiments, the melting temperature of the adhesive layer **30** may be greater than about 260° C.

[0050] Referring to FIG. **7**C, a pretreated or preformed conductive post **80** is disposed on the adhesive layer **30**. The disposing of the pretreated conductive post **80** may include pressing the pretreated conductive post **80** into the adhesive layer **30** before the curing of the adhesive layer **30**. In some embodiments, the conductive post **80** include, for example, Cu, other metal, a metal alloy, other conductive material, or a combination of two or more thereof. The lower surface **801***b* of the conductive post **80** is rough. The lower surface **801***b* of the conductive post **80** may have some small voids thereon. At least a portion of the adhesive layer **30** may be filled into the voids of the lower surface **801***b*. A relatively smaller sized conductive filler of the adhesive layer **30** may be fitted into the uneven surface **801***b* of the conductive post **80** to mitigate the void issue. The relatively great roughness of the lower surface **801***b* of the conductive post **80** may enhance the adhesion and conductivity between the lower surface **801***b* and the adhesive layer **30**. A relatively greater sized conductive filler of the adhesive layer **30** is in contact with the lower surface **801***b* and the pad **60**. After disposing the conductive post **80** into the adhesive layer **30**, the adhesive layer **30** is cured (e.g., hardened). Next, the semiconductor device package **1** of FIG. **1** is obtained. A portion of the lateral surface **801***s* of the conductive post **80** is surrounded by the adhesive layer **30**, which have a relatively high melting point, and may fix or support conductive post **80** during thermal cycles in a process of manufacturing the semiconductor device package **1**.

[0051] As used herein, the terms "approximately," "substantially," "substantial" and "about" are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a variation of less than or equal to ±10% of the numerical value, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%. Thus, the term "approximately equal" in reference to two values can refer to a ratio of the two values being within a range between and inclusive of 0.9 and 1.1.

[0052] Additionally, amounts, ratios, and other numerical values are sometimes presented herein in a range format. It is to be understood that such range format is used for convenience and brevity and should be understood flexibly to include numerical values explicitly specified as limits of a range, but also to include all individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly specified.

[0053] As used herein, the singular terms "a," "an," and "the" may include plural referents unless the context clearly dictates otherwise. In the description of some embodiments, a component provided "on" or "over" another component can encompass cases where the former component is directly on (e.g., in physical contact with) the latter component, as well as cases where one or more intervening components are located between the former component and the latter component.

[0054] While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations do not limit the present disclosure. It should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present disclosure as defined by the appended claims. The illustrations may not necessarily be drawn to scale. There may be distinctions between the artistic renditions in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. There may be other embodiments of the present disclosure which are not specifically illustrated. The specification and drawings are to be regarded as illustrative rather than restrictive. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective, spirit and scope of the present disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein have been described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations of the present disclosure.

What is claimed is:

1. A semiconductor device package, comprising:

a carrier;

a conductive post including two side surfaces; and

a first adhesive layer including an adhesive and conductive particles within the adhesive; and

an insulating layer, wherein, in a cross-section view, the insulating layer contacts a first portion of one of the two side surfaces of the conductive post, and the first adhesive layer contacts a second portion of the one of the two side surfaces of the conductive post,

wherein the first adhesive layer is disposed between the carrier and the conductive post, wherein, in the cross-section view, two opposite side surfaces of the conductive post are substantially parallel to each other and substantially perpendicular to an upper surface of the carrier.

2. The semiconductor device package of claim **1**, wherein, in the cross-section view, a length of the second portion of the one of the two side surfaces of the conductive post is less than a length of the first portion of the one of the two side surfaces of the conductive post.

3. The semiconductor device package of claim **2**, wherein a surface of the second portion of the one of the two side surfaces of the conductive post is continuous and coplanar with a surface of the first portion the one of the two side surfaces of the conductive post, and wherein a portion of the one of the two side surfaces of the conductive post is exposed from the first adhesive layer.

4. The semiconductor device package of claim **3**, wherein the surface of the second portion of the one of the two side surfaces of the conductive post is orthogonal to a bottom surface of the conductive post and wherein a step structure defined by the surface of the second portion and the bottom surface of the conductive post is surrounded by the first adhesive layer.

5. The semiconductor device package of claim **4**, further comprising a conductive pad, and wherein the step structure and the conductive pad are separated by the first adhesive layer.

6. The semiconductor device package of claim **4**, further comprising an electronic component disposed on the carrier, wherein the conductive post is electrically connected with a second conductive post through the electronic component.

7. A semiconductor device package, comprising:

a substrate;

a first conductive post disposed over the substrate;

a second conductive post disposed over the substrate and electrically connected with the first conductive post by the substrate;

a first adhesive layer electrically connecting the first conductive post to the substrate;

an electronic component disposed over the substrate and not vertically overlapped with the first conductive post and the second conductive post;

a conductive pad disposed over the substrate; and

a dielectric layer disposed over the substrate, wherein the conductive pad is disposed within an opening of the dielectric layer, wherein the first adhesive layer has an inclined sidewall connecting a side surface of the first conductive post and a side surface of the dielectric layer within the opening.

8. The semiconductor device package of claim **7**, wherein, in a cross-section view, the first adhesive layer covers two side surfaces of the first conductive post, wherein the two side surfaces are opposite to each other, and wherein a distance between a bottom surface of the first conductive post and an upper surface of the conductive pad is less than a thickness of the conductive pad.

9. The semiconductor device package of claim **7**, wherein a top surface of the first conductive post is at a first elevation level greater than a second elevation level of a top surface of the electronic component and a bottom surface of the first conductive post is at a third elevation level lower than a fourth elevation level of a bottom surface of the electronic component.

10. The semiconductor device package of claim **7**, further comprising an insulating layer surrounding the inclined sidewall of the first adhesive layer and a top surface of the electronic component.

11. The semiconductor device package of claim **10**, wherein the electronic component includes an active component and wherein the insulating layer is disposed between the electronic component and the substrate.

12. The semiconductor device package of claim **7**, wherein a roughness of a bottom surface of the first conductive post is greater than a roughness of the side surface of the first conductive post.

13. The semiconductor device package of claim **10**, wherein, in a cross-section view, the insulating layer contacts a first portion of the side surface of the first conductive post, and the first adhesive layer contacts a second portion of the side surface of the first conductive post.

14. The semiconductor device package of claim **13**, wherein, in the cross-section view, a length of the second portion is less than a length of the first portion.

15. A semiconductor device package, comprising:

a substrate having an upper surface;

a conductive pad disposed over the substrate;

a first conductive post disposed over the substrate and tilted with respect to the upper surface of the substrate; and

a first adhesive layer over the substrate, and wherein a first portion of a bottom of the first conductive post is in contact with the conductive pad, and a second portion of the bottom of the first conductive post is not in contact with the conductive pad, and the first adhesive layer is filled between the second portion of the bottom of the first conductive post and a top surface of the conductive pad.

16. The semiconductor device package of claim **15**, wherein a width of the first conductive post is greater than or equal to a half of a width of the conductive pad in a cross-sectional view, and wherein the conductive pad is disposed between the first conductive post and the substrate.

17. The semiconductor device package of claim **15**, further comprising an insulating layer, wherein, in a cross-section view, the insulating layer contacts a first portion of a side surface of the first conductive post, and the first adhesive layer contacts a second portion of the side surface of the first conductive post.

18. The semiconductor device package of claim **17**, wherein, in the cross-section view, a length of the second portion of the second portion of the side surface of the first conductive post is less than a length of the first portion the of the side surface of the first conductive post.

19. The semiconductor device package of claim **18**, wherein a surface of the second portion of the side surface of the first conductive post is continuous and coplanar with a surface of the first portion the side surface of the first conductive post.

20. The semiconductor device package of claim **19**, wherein the surface of the second portion of the side surface of the first conductive post is orthogonal to the bottom surface of the first conductive post, and wherein a step structure defined by the surface of the second portion and the bottom surface of the first conductive post is surrounded by the first adhesive layer.

* * * * *